

Evan Collins

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EDUCATION

Massachusetts Institute of Technology Doctor of Philosophy (PhD) Candidate in Biological Engineering Advised by Robert Langer ScD, Daniel Anderson PhD GPA: 5.00/5.00	2022 – present Cambridge, MA
Yale University Master of Science (MS) in Biomedical Engineering Bachelor of Science (BS) in Biomedical Engineering GPA: 3.96/4.00; Highest in major; High Honors with Distinction	2018 – 2022 New Haven, CT
<i>Non-degree education:</i> Tsinghua University Inter-University Program for Chinese Language Study (IUP 汉语培训项目)	2020 Beijing, China (remote)

CORE WORK EXPERIENCE

Massachusetts Institute of Technology, Koch Institute for Integrative Cancer Research Doctoral Researcher	2022 – present Cambridge, MA
Yale School of Medicine Student Investigator	2018 – 2022 New Haven, CT
Simplex Sciences Chief Operating Officer & Chief Science Officer	2019 – 2022 New Haven, CT
D. E. Shaw & Co. Risk Analyst Intern	2021 New York, NY
Click Therapeutics Data Scientist	2020 – 2021 New York, NY

AWARDS

- Sigma Xi Scientific Research Honor Society (2026)
- MIT Wishnok Prize for Best Talk (2025)
- MIT Prize for Open Data Honorable Mention (2025)
- MIT Jameel Clinic Fellowship for Machine Learning in Health (2022)
- Yale D. Allan Bromley Prize for Highest GPA in Biomedical Engineering (2022)
- Yale SEAS Belle and Carl Morse Scholarship (2021)
- D. E. Shaw Nexus Fellowship (2021)
- Yale Richard U. Light Fellowship for Chinese Studies (2020)
- Tau Beta Pi Engineering Honor Society (2020 – 2022)
- Lockheed Martin STEM Scholarship (2019 – 2022)
- Yale Summer Research Fellowship (2019)
- Intel International Science and Engineering Fair Third Place Prize (2018)
- National Merit Finalist and Scholarship (2018)
- Valedictorian of Ridgeview High School (2018)
- Florida Sunshine State Scholar (2017)

TEACHING

Massachusetts Institute of Technology Machine Learning for Molecular Engineering (3.C01/3.C51/10.C01/20.C01/20.C51) Course Content Developer	2024 Cambridge, MA
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Massachusetts Institute of Technology
Principles of Molecular Bioengineering (20.420[J])
Teaching Assistant

2023
Cambridge, MA

Yale University
Physiological Systems (BENG 350)
Peer Tutor

2020
New Haven, CT

Yale University
Ordinary and Partial Differential Equations (ENAS 194)
Peer Tutor

2020
New Haven, CT

PUBLICATIONS

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- "Quantitative Cryogenic Orbitrap Secondary Ion Mass Spectrometry (Q-Cryo-OrbiSIMS) demonstrated for lipid nanoparticles (LNPs)" 2026
Roberts J, Kotowska A, Aguilar-Bejarano E, **Collins E**, Witten J, Howard A, Trindade G, Gilmore I, Hook A, Winkler D, Alvey K, Figueredo G, Scurr D J, Anderson G D, Alexander M R
ChemRxiv. Under review.
<https://doi.org/10.26434/chemrxiv.15001097/v1>
- "Translating a review of language function to the parcellation of a high-resolution brain atlas" 2026
Watson E, **Collins E**, Chishti O, Bose-Roy R, McGrath H, Segal N A, Lehman R, Sivaraju A, Zaveri H P, Spencer D D
Epilepsia
<https://doi.org/10.1002/epi.70229>
- "Lipid Nanoparticle Database towards structure-function modeling and data-driven design for nucleic acid delivery" 2026
Collins E*, Ji J*, Kim SG*, Witten J, Kim S, Zhu R, Park P, Jung M, Park A, Manan R S, Rudra A, Jeang W J, Langer R, Anderson D G, Im W
Nature Communications
<https://doi.org/10.1038/s41467-026-68818-1>
- "Multitask Language Mapping to Visualize the Spatial Configuration of Polyfunctional Language Cortex" 2025
Chishti O, **Collins E**, McGrath H, Zhang T, Quraishi I H, Hirsch L J, Benjamin C, Damisah E C, Zaveri H P, Spencer D D, Sivaraju A
Neurology
<https://doi.org/10.1212/WNL.0000000000214256>
- "Self-driving labs for biotechnology" 2025
Collins E, Langer R, Anderson D G
Nature Computational Science
<https://doi.org/10.1038/s43588-025-00885-8>
- "Yale Brain Atlas to interactively explore multimodal structural and functional neuroimaging data" 2025
Collins E, Chishti O, McGrath H, Obaid S, King A, Qiu E, Gabriel E, Shen X, Arora J, Papademetris X, Constable T R, Spencer D D, Zaveri H P
Frontiers in Network Physiology
<https://doi.org/10.3389/fnetp.2025.1585019>
- "Antifouling Immunomodulatory Copolymer Architectures that Inhibit the Fibrosis of Implants" 2025
Jeang W J*, Wong B M*, Zhao Y, Manan R, Jiang A L, Bose S, **Collins E**, McMullen P, Rosenboom J, Lathwal S, Langer R, Anderson D G
Advanced Materials
<https://doi.org/10.1002/adma.202414743>

- "Artificial intelligence-guided design of lipid nanoparticles for pulmonary gene therapy" 2024
Witten J*, Raji I*, Manan R S*, Beyer E, Bartlett S, Tang Y, Ebadi M, Lei J, Nguyen D, Oladimeji F, Jiang A Y, MacDonald E, Hu Y, Mughal H, Self A, **Collins E**, Yan Z, Engelhardt J F, Langer R, Anderson D G
Nature Biotechnology
<https://doi.org/10.1038/s41587-024-02490-y>
- "Mapping the structure-function relationship along macroscale gradients in the human brain" 2024
Collins E, Chishti O, Obaid S, King A, McGrath H, Shen X, Arora J, Papademetris X, Constable T R, Spencer D D, Zaveri H P
Nature Communications
<https://doi.org/10.1038/s41467-024-51395-6>
- "Systematic 1 Hz Direct Electrical Stimulation for Seizure Induction: A Reliable Method for Localizing Seizure Onset Zone and Predicting Seizure Freedom" 2024
Sivaraju A, Quraishi I, **Collins E**, McGrath H, Ramos A, Turk-Browne N, Zaveri H P, Damisah E, Spencer D D, Hirsch L J
Brain Stimulation
<https://doi.org/10.1016/j.brs.2024.03.011>
- "High-resolution cortical parcellation based on robust brain landmarks for precise localization of multimodal data" 2022
McGrath H, Zaveri H P, **Collins E**, Jafar T, Chishti O, Obaid S, Ksendzovsky A, Wu K, Papademetris X, Spencer D D
Scientific Reports
<https://doi.org/10.1038/s41598-022-21543-3>
- "The Implied Truth Effect: Attaching Warnings to a Subset of Fake News Headlines Increases Perceived Accuracy of Headlines Without Warnings" 2020
Pennycook G, Bear A, **Collins E**, & Rand D G
Management Science
<https://doi.org/10.1287/mnsc.2019.3478>

CONFERENCES

- Collins E**. "Data-driven approaches for understanding and improving mRNA lipid nanoparticles". Talk at *MIT Koch Institute for Integrative Cancer Research Retreat* (2025).
- Collins E**, Ji J, Kim SG, Witten J, Kim S, Zhu R, Park P, Jung M, Park A, Manan R S, Rudra A, Jeang W J, Im W, Langer R, Anderson D G. "LNPDDB: structure-function dataset of lipid nanoparticles to advance data-driven design for nucleic acid delivery". Poster at *MIT Molecular Machine Learning Conference (MoML)* (2025).
- Park P, **Collins E**, Kim S, Im W. "Improving Martini 3 Coarse-Grained Models for Ionizable Lipids in LNP Simulations". Poster at *American Chemical Society Fall 2025 conference* (2025).
- Ji J, **Collins E**, Kim SG, Kim S, Im W. "LNPDDB: structure-function database of lipid nanoparticles to advance data-driven design for nucleic acid delivery". Poster at *American Chemical Society Fall 2025 conference* (2025).
- Collins E**, Chen M, Langer R, Anderson D G. Poster at *MIT Department of Biological Engineering Retreat* (2024).
- Watson E, **Collins E**, Chishti O, McGrath H, Sivaraju A, Zaveri H P, Spencer D D. "Yale Brain Atlas Parcellation of Overlapping Multimodal Functions in the Human Temporal Lobe". Poster at *American Epilepsy Society conference* (2024).
- King A, Obaid S, Pu K, Chishti O, **Collins E**, McGrath H, King-Stephens D, Duckrow R, Spencer D D, Zaveri H P. "Use of the Yale Brain Atlas to Determine Signatures of Cortical Thinning". Poster at *American Epilepsy Society conference* (2024).
- Pu K, Chen M, Chen C, King A, **Collins E**, Spencer D D, Zaveri H P. "PET Hypometabolism is Associated with Epileptogenesis at Multiple Spatial Resolutions Ranging from the Hemisphere of Seizure Onset to the Seizure Onset Area". Poster at *American Epilepsy Society conference* (2023).
- Watson E, **Collins E**, Chishti O, McGrath H, Sivaraju A, Zaveri H P, Spencer D D. "The optimization of a high-resolution brain atlas as a tool for surgical planning by translating a review of language function to parcellation". Poster at *American Epilepsy Society conference* (2023).

Collins E, Chishti O, Obaid S, McGrath H, King A, Spencer D D, Zaveri H P. “Large-scale analysis of the relationship between structure and function in the human cortex”. Poster at *American Society of Neuroradiology Annual Meeting* (2023).

Collins E, Chishti O, Obaid S, McGrath H, Jafar T, King A, Zaveri H P, Spencer D D. “Anatomical brain atlas for structure-function coupling: predicting surgical outcome from seizure spread and analyzing large-scale fMRI-connectome associations”. Poster and presentation at *American Epilepsy Society conference* (2022).

Collins E, McGrath H, Zaveri H P, Papademetris X, Wu K, Spencer D D. “Systematic Parcellation of the Human Cortex for Comparison of Multimodal Neuroimaging Data”. Poster at *American Epilepsy Society conference* (2021).

Jafar T, McGrath H, Ksendzovsky A, Zaveri H, Farooque P, **Collins E**, Sivaraju A, Damisah E, Papademetris X, Spencer D D. “A multimodal cortical atlas for clinical decision making and function-structure hypotheses in epilepsy surgery”. Poster at *Society for Neuroscience conference* (2021).

ACADEMIC SERVICE

Mentorship

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| • Massachusetts Institute of Technology, Undergraduate thesis research | 2023-present |
| • Yale University, Undergraduate thesis research | 2021-present |

Public engagement and science communication

Press interviews for *Nature*, *YaleNews*

Peer review

Contributed reviews to *Brain Structure and Function*, *Brain and Cognition*, *Epilepsy & Behavior*, *Molecular Therapy – Nucleic Acids*, *Nature Biotechnology*, *Nature Reviews Materials*, *Technology in Society*